

Sample AI Contribution Statement

Student Name(s): _____

Course: COEN 448/6761 – Software Testing and Validation

Assignment: Integration Testing and Test Coverage

1. AI Tool Utilization Overview

In this assignment, AI (e.g., Gemini, ChatGPT, Claude) was utilized to support the following tasks:

- **Environment Configuration:** AI assisted in generating the initial `docker-compose.yml` structure to network the User and Order microservices with RabbitMQ.
- **Test Case Expansion:** AI was used to draft detailed steps for **Test Case 3 (Event Propagation)** and **Test Case 4 (Strangler Pattern)** based on the provided high-level objectives.
- **Scripting Support:** AI provided the boilerplate code for the Python `requests` script used to validate the `SP$` percentage weight-based routing in the Kong API Gateway.

2. Prompt Log and Iteration (Illustrating Cases)

To demonstrate the "Human-in-the-Loop" process, below are specific examples of how AI outputs were verified or corrected:

AI Output Provided	Verification/Correction Performed by Student
Generated a Docker config using RabbitMQ port 5672.	Corrected the port as per the specific deployment environment.
Suggested a simple if-else for the Strangler Pattern.	Refined the logic to ensure the P value was pulled from an external configuration file rather than being hardcoded.
Generated a generic User JSON schema.	Modified the schema to strictly adhere to the <code>user_schema.json</code> requirements, ensuring fields for email and delivery address were mandatory.